Insurance Application

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Outline

Introduction/Requirements

Technologies Used

Project Flow

4+1 Diagram

API's Used

Introduction

- Building an individual and family insurance application just by having APIs.
- Provides two types of insurance:
  1. Life
  2. Dental
- The costing of the plan is dependent on multiple parameters, such as:
  - The number of dependents enrolled
  - The age of the primary subscriber

Introduction Conti...

- There are multiple roles of users, such as End User and Admin.
- The insurance application has to follow a workflow where:
  - Once submitted and passing all the rules set by the admin, it gets auto-approved.
- All the submitted applications need to be reviewed at periodic intervals (e.g., every 4 hours).
  - Use a scheduler or Spring Batch job for this task.

Note:

- All the functionality of the application, such as login, registration, and listing of available products, etc., should be shown via REST calls by providing necessary parameters in the payload. No UI rendering is needed.

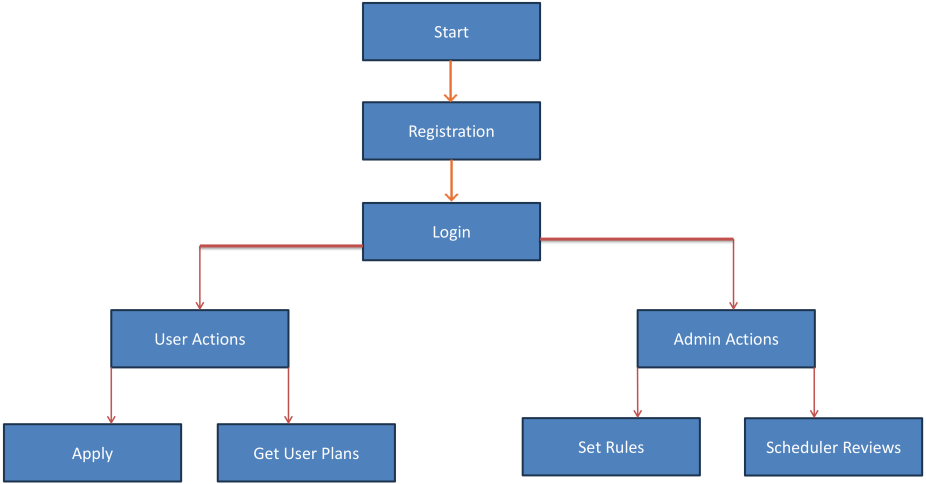
Technologies used

- Java/J2EE
- Rest Services with JWT security
- My-SQL
- Scheduler
- For security and best practices (Global exception handling, validations, Logger)
- Added validation in two layers
  1. Database layer (in the entities)
  2. API layer (in the RequestBody)

Project Flow

- Step 1: User registers or logs in.
- Step 2: User selects the insurance type.
- Step 3: Application is validated and processed.
- Step 4: Admin rules determine approval or rejection.

Project Flow Diagram



4+1 Architectural Views

Overview:

Illustrating the logical, development, process, and physical views of the system.

4+1 View - Class Diagram

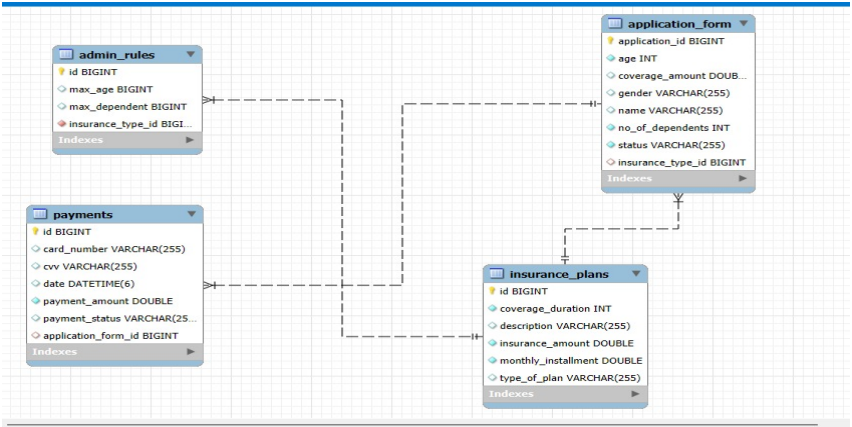


Figure 1: Class Diagram View

4+1 View - Activity Diagram

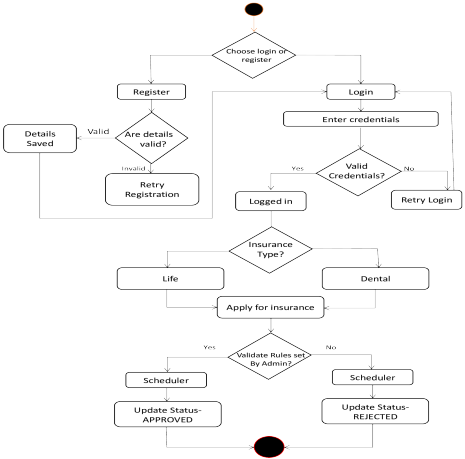


Figure 2: Activity Diagram View

4+1 View - Use Case Diagram

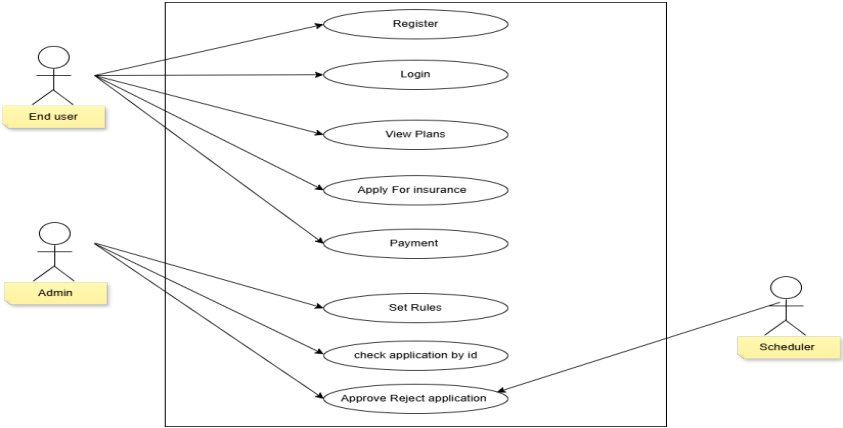


Figure 3: Use Case Diagram View

API's Used

- **AdminRulesController**: *SetRules*
- **ApplicationController**: *apply*, *getById*
- **Type_of_InsuranceController**: *getInsurancePlans*
- **UserController**: *Login*, *Register*
- **ProcessPayment**

User API: Registration and Login

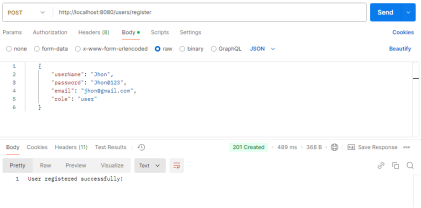
Register

Allows users to register with necessary credentials and details.

Login

Authenticates users and provides access tokens for secure sessions.

User API: User Registration



	id	password	user_name	email	role
	1	$2a$12$Xwi1dv5Kkz7NRcKUmCKLNeVkNv0rN0...	Vansh	vansh@gmail.com	user
	2	$2a$12$4qntSLKMXDRSTPcRRuRaDee9hQcSk/P...	Rohan	rohan@gmail.com	user
	3	$2a$12$vCnM3XXIa5280EcnkzG4BO.AdT.j7rX3...	Raghav	raghav@gmail.com	user
	4	$2a$12$3LEwUl8Vm53MkEyxfOueQuEa.7WLUI...	Anshika	anshika@gmail.com	admin
▶	9	$2a$12$9ICUEt27762AY2vebJJteKoqIJuJny.zsg...	Jhon	jhon@gmail.com	user
✱	NULL	NULL	NULL	NULL	NULL

Figure 4: User Registration

User API: User Login

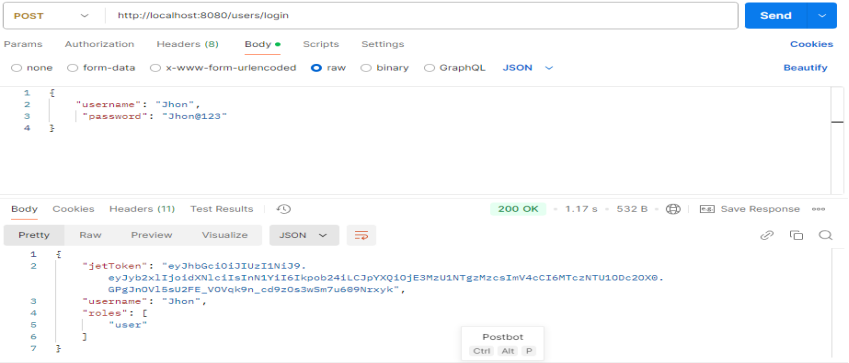


Figure 5: User Login

Admin API: Rule Management

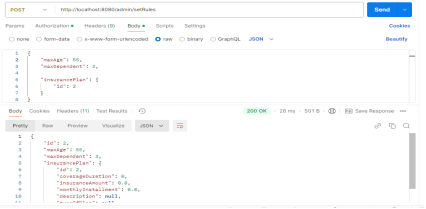
Set Rules

Enables administrators to define and manage application rules and policies.

Find by ID

Retrieves specific admin-related data or user profiles using unique identifiers.

Admin API: Set rules



	id	max_age	max_dependent	insurance_type_id
▶	1	60	4	1
	2	55	3	2
✱	NULL	NULL	NULL	NULL

Figure 6: Set rules

Admin API: Find by id

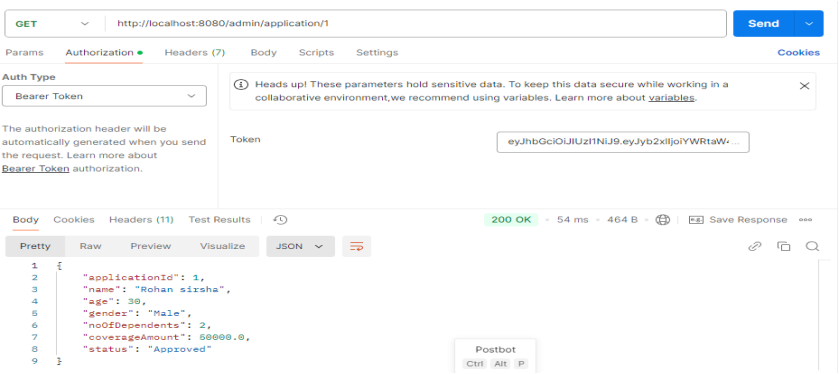


Figure 7: Find by id

Application API: User and Insurance Management

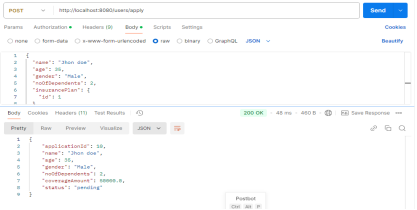
User Application

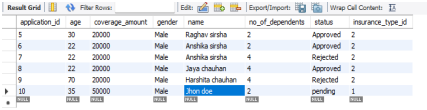
Manages user-submitted applications and their statuses.

Type of Insurance

Provides information and management options for different insurance types.

Application API: User Application





Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

application_id	age	coverage_amount	gender	name	no_of_dependents	status	insurance_type_id
5	30	20000	Male	Raghav sirsha	2	Approved	2
6	22	20000	Male	Anshika sirsha	2	Approved	2
7	22	20000	Male	Anshika sirsha	4	Rejected	2
8	22	20000	Male	Jaya chauhan	4	Approved	2
9	70	20000	Male	Harshita chauhan	4	Rejected	2
10	35	50000	Male	John doe	2	pending	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 8: User Application

Application API: Type of Insurance

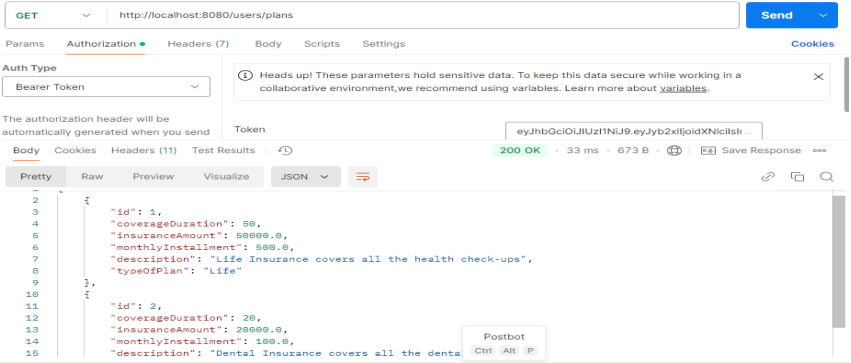
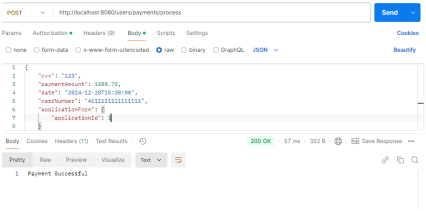
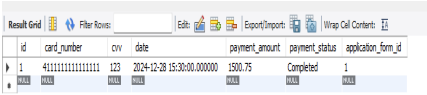


Figure 9: Type of Insurance

Dummy Payment System





id	card_number	cvv	date	payment_amount	payment_status	application_form_id
1	4111111111111111	123	2024-12-28 15:30:00.000000	1500.75	Completed	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 10: Dummy Payment System

Scheduler Review API

Scheduler Review

Automates periodic reviews and schedules for user or application evaluations.

Scheduler Review API

```
Scheduler started!
Hibernate: select af1_0.application_id,af1_0.coverage_amo
Hibernate: select toi1_0.id,toi1_0.insurance_amount,toi1_
Found: 1 pending applications
Hibernate: select ar1_0.id,ar1_0.max_age,ar1_0.max_depend
Application 1 APPROVED!!
Hibernate: select af1_0.application_id,af1_0.coverage_amo
Hibernate: update application_forms set coverage_amount=?
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

application_id	age	coverage_amount	gender	name	no_of_dependents	status	insurance_type_id
5	30	20000	Male	Raghav sirsha	2	Approved	2
6	22	20000	Male	Anshika sirsha	2	Approved	2
7	22	20000	Male	Anshika sirsha	4	Rejected	2
8	22	20000	Male	Jaya chauhan	4	Approved	2
9	70	20000	Male	Harshita chauhan	4	Rejected	2
10	35	50000	Male	Jhon doe	2	Approved	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 11: Scheduler Review

Future Aspects

- Add a user-friendly interface using Angular.
- Add more APIs and features like editing the application.
- Add more users, such as underwriters.

Thank You!